
Course Name: Machine Learning for Earth System Sciences

Week 04 - Assignment 04 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10

Total marks: 10x1= 10

QUESTION 1

Which of the following statements about the Intertropical Convergence Zone (ITCZ) and the Indian Summer Monsoon is correct?

- (A) The ITCZ remains stationary throughout the year.
- (B) The seasonal northward migration of the ITCZ contributes to the establishment of the monsoon trough and enhanced convection over India.
- (C) The ITCZ primarily influences winter rainfall over India.
- (D) The ITCZ suppresses the transport of moisture from the Arabian Sea.

Answer: B

Detailed Solution:

ITCZ stands for Inter-Tropical Convergence Zone. The ITCZ shifts northward in the Northern Hemisphere summer, helping establish the monsoon trough over India and enhancing convection and rainfall.

QUESTION 2

A climate scientist needs to generate high-resolution projections of future rainfall over the Western Ghats. Limited computational resources are available, but several decades of reliable historical observations exist to establish relationships between large-scale climate variables and local rainfall. Which downscaling approach is most appropriate?

- (A) Dynamical downscaling using a Regional Climate Model (RCM)
- (B) Statistical downscaling using empirical relationships
- (C) Ensemble Kalman Filtering
- (D) Principal Component Analysis

Answer: B

Course Name: Machine Learning for Earth System Sciences

Week 04 - Assignment 04 - (Jul-2026)

Detailed Solution:

Dynamic Downscaling requires running a high-resolution Regional Climate Model (RCM) requires high-performance computing (HPC) cluster. Statistical downscaling, conversely, uses mathematical, statistical, or machine learning models (that run in a fraction of the time with minimal computing power). In addition, historic data is also available for training of ML models.

QUESTION 3

In the study [Misra S, Sarkar S, Mitra P. *Statistical downscaling of precipitation using long short-term memory recurrent neural networks. Theoretical and applied climatology. 2018 Nov;134(3):1179-96.*], the model used for statistical downscaling is

- (A) RNN and the results have been compared with LSTM based methods.
- (B) RNN-LSTM and the results have been compared with Regression based and Deep Neural Network based methods.
- (C) LSTM and the results have been compared with RNN based methods.
- (D) RNN-LSTM and the results have been compared with HMM based methods.

Answer: B

Detailed Solution:

RNN-LSTM and compared with regression-based methods refer to the research paper mentioned in the question

Course Name: Machine Learning for Earth System Sciences

Week 04 - Assignment 04 - (Jul-2026)

QUESTION 4

In Markov Random Field based Anomaly detection, the edge potential functions between latent variables indicates that:

- (A) The anomaly events are not spatio-temporally coherent.
- (B) the anomaly events are spatio-temporally coherent.
- (C) the anomaly events are spatial coherent but not temporal coherent.
- (D) the anomaly events are temporal coherent but not spatial coherent.

Answer: B

Detailed Solution:

In MRF based Anomaly detection the edge potential functions between latent variables indicate that the anomaly events are spatio-temporal coherent.

QUESTION 5

The Raman Research Laboratory has gathered data for a set of climate variables that are correlated with one another and have nonlinear dependencies on one another. The data was collected for 50 years with a temporal resolution of 24 hours. They have employed a complex spatio-temporal technique for causality detection using this data and have faced few process-level challenges in this task. Which of the following will not be considered as a process-level challenge for this study?

- (A) Autocorrelation
- (B) Time delays
- (C) Nonlinear dependencies
- (D) Sample Size

Answer: D

Detailed Solution:

Autocorrelation, time delays and nonlinear dependencies are process level challenges whereas sample size is a computational challenge.

Course Name: Machine Learning for Earth System Sciences

Week 04 - Assignment 04 - (Jul-2026)

QUESTION 6

Identify the False statement

- (A) Causal discovery is responsible for analysing and creating models that illustrate the relationships inherent in the data.
- (B) Causal inference aims to study the possible effects of altering a given system.
- (C) PC algorithm identifies causes of each variable with confounders.
- (D) Granger causality works well in the presence of confounders and contemporaneous relations.

Answer: D

Detailed Solution:

Granger causality does not work well in the presence of confounders and contemporaneous relations.

QUESTION 7

In the study for the identification of the predictors of Indian monsoon [Saha M, Mitra P, Nanjundiah RS. *Autoencoder-based identification of predictors of Indian monsoon. Meteorology and Atmospheric Physics.* 2016 Oct;128(5):613-28.], the primary task of autoencoder is

- (A) Identifying causal relationships
- (B) Identifying the temporal autocorrelations of rainfall data
- (C) Dimension reduction
- (D) Downscaling the outputs from a GCM model to Indian subcontinent regional model

Answer: C

Detailed Solution:

Course Name: Machine Learning for Earth System Sciences

Week 04 - Assignment 04 - (Jul-2026)

Autoencoders have been used in the referenced paper for dimension reduction of the monsoon rainfall data.

QUESTION 8

Which of the following Earth system applications is best suited to a U-Net architecture because it requires assigning a class label to every pixel in a satellite image?

- (A) Flood extent mapping from satellite imagery
- (B) Cyclone track prediction from historical observations
- (C) Detection and localization of buildings damaged after an earthquake from satellite imagery using bounding boxes
- (D) Compression of climate model outputs

Answer: A

Detailed Solution:

UNet was designed primarily for semantic image segmentation. Cyclone track prediction requires modelling temporal sequential trajectories for which RNNs (LSTMs) and Transformers are suited. For detection of buildings object detection models like YOLO are more suited. Compression tasks are suited for Autoencoders.

QUESTION 9

Identify the True statement

- (A) Granger causality detection is a conditional independence-based technique, but PC algorithm is a conditional dependence-based technique.
- (B) Granger causality detection is a conditional dependence-based technique, but PC algorithm is a conditional independence-based technique.
- (C) Both Granger causality detection and PC algorithm are conditional independence-based techniques.
- (D) Both Granger causality detection and PC algorithm are conditional dependence-based techniques.

Course Name: Machine Learning for Earth System Sciences

Week 04 - Assignment 04 - (Jul-2026)

Answer: B

Detailed Solution:

Granger causality detection is a conditional dependence-based technique, but the PC algorithm is a conditional independence-based technique.

QUESTION 10

In the study [Mitra A, Seshadri AK. *Detection of spatiotemporally coherent rainfall anomalies using Markov Random Fields. Computers & geosciences. 2019 Jan 1;122:45-53*], Gibbs sampling has been used for

- (A) Estimation of optimal values of latent state variables
- (B) Statistical Downscaling of rainfall data
- (C) Maximum likelihood estimation
- (D) Identification of the spatio-temporal extended rainfall anomalies

Answer: A

Detailed Solution:

Estimation of optimal values of latent state variables, Refer to the research paper